

Table 6.1 Summary Statistics

1927–1960	SP500	HML	LTR
Average	1.00%	0.36%	0.26%
Variance	0.005	0.002	0.000
Std. Dev.	7.04%	4.42%	1.26%
Correlation	SP500	HML	LTR
SP500	100.00%	58.93%	5.64%
HML	58.93%	100.00%	4.03%
LTR	5.64%	4.03%	100.00%
1961–2013	SP500	HML	LTR
Average	0.91%	0.40%	0.59%
Variance	0.002	0.001	0.001
Std. Dev.	4.32%	2.83%	2.36%
Correlation	SP500	HML	LTR
SP500	100.00%	–26.41%	16.91%
HML	–26.41%	100.00%	0.44%
LTR	16.91%	0.44%	100.00%

For many investors, the correlation matrix doesn't factor into asset allocation decisions, but for the herds of professional investors trained in the art of modern portfolio management, correlation matrix estimates matter a great deal. A stable correlation matrix is required if investors want to get reliable estimates for how to make allocations into their portfolio. Yet, as the evidence shows, correlation matrixes, even when calculated based on long time frames, can be highly unstable over discrete periods!

Let's take this concept a step further and look at two ways to create a portfolio using the methods discussed. [Table 6.2](#) shows the recommended portfolio weights using two methods, the tangency portfolio method (maximize risk-adjusted return) and the minimum variance method (minimize risk). Both methods will use our three asset classes: the S&P 500, the long/short value hedge fund, and long bonds. The tangency and